

Road Scene Content Analysis for Driver Assistance and Autonomous Driving

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Abstract—This paper aims to develop a vision-based driver assistance system for scene awareness using video frames obtained from a dashboard camera. A saliency image map is devised with features pertinent to the driving scene. This saliency map mimics the human contour and motion sensitive visual perception by extracting spatial, spectral, and temporal information from the input frames and applying entropy driven image-context-feature data fusion. The resultant fusion output comprises high-level descriptors for still segment boundaries and non-stationary object appearance. Following the segmentation and foreground object detection stage, an adaptive maximum likelihood classifier selects road surface regions. The proposed scene driven vision system improves the driver's situational awareness by enabling adaptive road surface classification. As experimental results demonstrate, context-aware low-level to high-level information fusion based on human vision model produces superior segmentation, tracking, and classification results that lead to high-level abstraction of driving scene.

Index Terms—Road scene, content analysis, saliency map, entropy-driven context-feature fusion.

I. INTRODUCTION

SCENE understanding has found growing applications in intelligent transportation with driver assistance and autonomous driving systems. Presently, there are many automobile models with driver assistance systems such as parking assistance, lane departure warnings, emergency brake assistance, adaptive cruise control, etc. [1], leading toward the development of autonomous vehicles and transportation. These vehicles are equipped with various different sensors including global positioning system (GPS), radio detection and ranging (Radar), and light detection and ranging (LIDAR). Autonomous cars are being designed in a manner that they can communicate with other autonomous vehicles in their vicinity and, in turn, form underlying sensor networks [2].

Intelligent vehicles' research has started more than two decades ago and it has rapidly grown in the last fifteen years. Majority of intelligent vehicle research in the literature is in line with the current autonomous vehicle design trends, which involve multiple sensors and associated networks. Studies such as [3] and [4] provide frameworks for combining a variety of sensors such as LIDAR, radar, and GPS. Detection rate in drivers' surrounding can benefit from using a variety of

sensors in expense of higher costs as the number of built in sensors increases. These sensors usually work independently from each other and they generate warnings and notifications at different time instances. Wagh *et al.* [5] argue that some of these notifications may not be of much significance for the driver and proposes a sensor data fusion method that generates more informative warnings and notifications. Works such as [6] and [7] focus on sensor networks and multivehicle cooperative driving. Such approaches enable trajectory planning to avoid collision. However, researchers in [1] contemplate that, with vehicles and sensors being produced by various manufacturers, sensor interoperability will be a challenge for sensor networks. Some computer vision methods detect and track objects such as moving vehicles [8]–[11] with varying degrees of success. Others including [12] and [13] detect and track road lines, pedestrians [14] and traffic lights [15].

Substantial studies have been published on segmentation and scene understanding. However, few of them focus on feature fusion and incorporation of temporal attributes. For instance, the approach in [16] uses fusion of directional spatial features such as eigenvectors, gradients and temporal saliency for vehicle detection from air. In [15], optical flow and a pixel based similarity metric are utilized to segment road scenes. Work in [16] also utilizes optical flow for traffic analysis using a fixed camera directed at an intersection. Color clustering and local binary patterns as a texture descriptor are employed for road scene segmentation in [17]. The method presented in [18] outputs large feature vectors using motion boundary histograms to classify road scenes. Some of these methods such as the latter case would be inefficient for video processing due to the size of their feature space. The approach undertaken in [19] makes use of texture anisotropy as a feature for detecting road segments. They combine multiple pixels into large super pixels to reduce complexity.

Previous works explored combining pairs of these three information sources such as spatial and spectral [17], spatial and temporal [14] or spectral and temporal [15]. There are also methods that integrate texture and motion with color information which is a spatial-temporal spectral feature [20], [21]. These methods use structural tensors to extract texture and motion information and attempt to segment the image based on feature vectors obtained for all pixels. These are parametric models and they require a-priori information such as number of segments and class means. These requirements make unsupervised clustering a difficult task. In most cases, they can only segment the image into two regions. Furthermore, none of the previously applied fusion

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methods are based on a human perception model. The fusion result of our work emulates the visual stimulus formed by the retinal receptor layer in order to yield better segmentation and scene understanding by utilizing all three visual information cues: spatial, spectral and temporal.

Despite the promising progress in autonomous vehicular technology, humans are still largely in control of operating vehicles using their eyes to perceive the surrounding environment and to encapsulate the driving scenery through all visually available information as a whole entity. Therefore, a camera only system should be able to extract context from scenery in a similar manner without needing extra sensors, if it utilizes all available visual information.

Human eyes are able to cluster similar colors and intensities in spontaneous vision. Neurons connected to receptors that receive similar hue and intensity levels fire at the same rate and what captures our attention the most are the transitions in color and intensity. The regions with high contrast form the region boundaries or contours of an object. The importance of contours in human perception can be conceded from the tendency of drawing the contours first when asked to sketch an object. This phenomenon was first observed by Ernst Mach. According to his theory, the perceived image is the actual image minus the Laplacian of the image multiplied with a coefficient. This has a sharpening effect on the perceived image [22]. In fact, finding zero crossings of an image Laplacian is an effective edge detection method in image processing literature. Humans also detect textures and texture changes easily. One of the most significant works on texture segmentation [23] lists texture components as periodicity, directionality, and randomness. When our eyes detect small changes in contrast in a periodic, directional or random manner, we recognize that as a texture. Over a large region these contrast changes add up and become easy to recognize. Although texture properties are mainly spectral rather than spatial, our perception clusters textures the same way it clusters color and intensity, and finds boundaries between different textures. Boundaries of textures are added to our contour vision.

Another observational cue emphasized frequently in human vision is motion. There are tracking neurons in visual system that fire upon detection of motion. The faster the observed object moves the more exhibited these neurons become [22]. This explains why moving objects immediately catch our attention.

Detected contours from colors, intensity, texture, and motion are synthesized to become our visual stimulus. The combined output of these visual features forms a map showing where most of our attention focuses in a scene. Continuous long curves in this map depict the boundaries of regions, whereas blobs with high concentration of contours or motion define foreground objects.

With computer vision methods, similar to human eye, spectral (color, intensity) and spatial-spectral (texture) information can be extracted. Moreover, motion detection algorithms such as optical flow can find temporal changes similar to the tracker neurons in our eyes. When these data are combined using the image fusion methods mentioned earlier, the result is a saliency map [24] showing object and region contours as

well as objects in motion. This saliency map models the visual stimulus after synthesis of spatial, spectral and temporal information. By using the contours in the saliency map, better semantic information can be captured. In turn, visually distinct semantically meaningful regions in an image can be captured as a result of the gross overall segmentation of the scene being processed.

The proposed system is able to identify road surface and background as well as the foreground objects. It is a vision only approach as opposed to other methods that rely on other sensors such as LIDAR. Segmentation of context descriptive objects is achieved herein by combining low level spatial, spectral and temporal data obtained from video stream into higher level descriptors. An error minimizing method ensures high accuracy and robust detection even under adverse conditions such as very low SNR. Utilization of such camera only systems that uses all available visual information, simplifies the implementation of intelligent cars and driver assistance framework.

The remainder of this paper is organized as follows. In section II, details of the proposed information fusion system is explained. In section III, the results are presented. The last chapter contains conclusions and future work.

II. PROPOSED VISUAL INFORMATION FUSION METHOD

In computer science terms, incorporating data from multiple sources into a single output is called information fusion or data fusion [25], [26]. The purpose of data fusion is to generate information that surpasses the individual sources used for fusion. Image fusion methods find applications in military target detection and tracking [27], surveillance [28], [29], remote sensing [30], medical imaging [31], etc. Fusion needs a basis, which is a set of shared attributes used as a guide to merge the information content into one meaningful representation. Among other methods [25], fusion can be based on entropy, fuzzy logic, wavelets and principal component analysis results. Image fusion involves images obtained from two or more different sources.

In this study, fusion operation is carried out with images obtained from a single camera. Each video frame goes through three different analysis procedures, namely, spatial, spectral and temporal analysis. These analysis procedures produce various low level image features such as color clustering, local variance, texture energy, and optical flow. In order to produce an image with higher information content, these low-level features are combined in such a way that the images with higher information contents are assigned higher weights. A sample frame sequence used for feature extraction is given in Fig. 1.

A. Extraction of Spatial-Domain Information

Spatial features are obtained by analyzing the video frames based on color and intensity. The first step in extracting spatial information is by finding pixels with similar colors and clustering them. In order to group pixels of similar color into clusters, a fast clustering algorithm needs to be formulated. Although k-Means Clustering [32] is a commonly used method, selection



Fig. 1. Original image sequence.



Fig. 2. Color clustering and cluster contours.

of optimal number of clusters (k) requires multiple calls to this method, which increases computational workload. Instead, a fast converging histogram based thresholding method is adopted in this work. The selected clustering method is a variation of work presented in [33]. Having this method as a base, the Otsu threshold is iterated recursively over the image histogram in process. In this strategy, Otsu's method [34] finds an optimal threshold and divides the image into a dark and a light class pattern. The optimal threshold is selected in such a way that it minimizes the within the cluster deviation and maximizes the between class scatter value. Each time a histogram is separated into a lower and a higher histogram variable range, the Otsu algorithm is called on these lower and higher histograms and the segment is divided into further smaller segments until termination conditions occur. Results of color clustering applied to sample frame can be seen in Fig 2.

Local variance appears to be a strong contour descriptor. In regions where image intensity changes abruptly, local variance is much higher as compared to uniform regions. Although local variance is listed as a spatial feature, it also contains some texture information. In uniform image regions local variance is close to zero. In regions with texture, local variance has high value due to the variations in pixel intensities. In that sense, it can be considered as a hybrid feature but the contributions from edges are much higher than the contributions from texture. Fig.3 depicts the local variance of the sample scene.

B. Extraction of Spectral-Domain Information

A possible approach to localize different textures is to find different energy levels in local regions, which can be carried out via one of the methods for computing image energies. Transform energies are chosen in this research due to their easy and fast calculation. Parseval's theorem and energy relationship of cosine transform indicate that signal energy is preserved after discrete Fourier transform (DFT) and discrete cosine transform (DCT) [35]. Moreover, by separating the DC component, energy due to gray level intensities can be ignored and energy due to transitions can be detained. Fig. 4 shows the result of texture based gross overall road scene segmentation toward context manifestation.

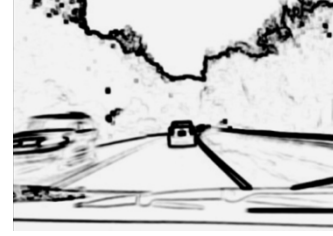


Fig. 3. Local variance.

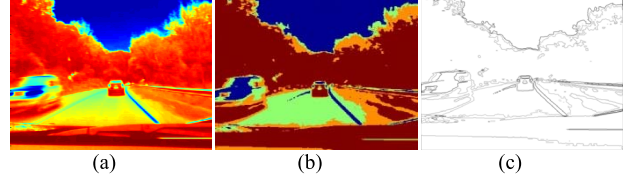


Fig. 4. Texture features from local cosine transform energy, clusters based on texture and their contours.

C. Extraction of Temporal Information With Optical Flow

Temporal analysis is performed by calculating the optical flow of video sequence. Optical flow is a powerful tool for detecting motion in a video. Regions containing moving objects in video frames can be identified with optical flow. In addition to motion detection, the direction and magnitude of the motion can be calculated with optical flow. Many methods have been proposed for computation of optical flow [36]–[38]. For real time performance constraint, we devise a method with low computational cost, which is similar to the approach for optical flow proposed in [39]. Figure 5 demonstrates the optical flow output of sample frames given in Fig. 1.

D. Information Fusion

The process of combining multiple sources of data into a more informative form is called data fusion. When these data sources are images, the process is called image data fusion or image fusion in short. There are various approaches for fusion [25]. In view of context aware scene partitioning in the domain of spatial, spectral and temporal image attributes, the approach undertaken in this research is feature level fusion. In selection of the fusion method, ideally the features with higher information must have more contribution to the fused output compared to other features with less information. In this work, starting with two images I_1 and I_2 , we combine the associated information of I_1 and I_2 by scaling one of them in such a way that the fusion outcome emphasizes their common information.

Let us assume that two inputs X and Y have common information and information exclusive to one of them (i.e., mutual information and conditional entropy). Starting with two sources, if we want to emphasize the redundant information in our fused output, we maximize mutual information, namely, $H_M(X, Y)$. If we want to emphasize information specific to individual sources, we need to maximize the union of the conditional entropies; i.e., $H(X|Y) \cup H(Y|X)$ where U is the union. This suggests that mutual information between images that represent different features should be maximized to find the best combination of those images.

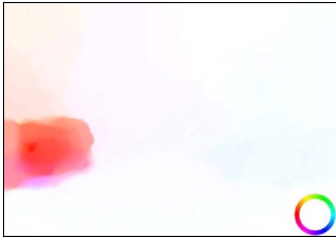


Fig. 5. Optical flow of the 2 frame sequence. Notice the color wheel indicates direction and saturation represents magnitude of motion.

Assume that there are two inputs with the histograms shown in Fig. 6. The fused image will be combination of the two:

$$I_f = aI_1 + bI_2 \quad (1)$$

Finding optimal I_f becomes equivalent to finding the ratio of a and b coefficients for I_1 and I_2 . Since the output will be normalized, this can be further reduced to:

$$I_f = \alpha I_1 + I_2 \quad (2)$$

Next, we need to select which of the two inputs should be scaled. Mutual information will be maximum when the two images are most similar, which, in most cases, means when their histograms are most similar. In the selected case without losing generality, I_1 has more variance compared to I_2 . Therefore, we can pick $\alpha < 1$ and use it to scale I_1 and shrink its histogram or we can pick $\alpha > 1$ and use it to stretch the histogram of I_2 . If we use $\alpha > 1$, then some of the pixels in I_2 will be scaled beyond the maximum brightness level allowed. We are going to either lose those pixels or limit them to the maximum attainable value (e.g., 1 or 255). Either case will introduce non-linearity and result in information loss. Thus, we should select $\alpha < 1$ and scale I_1 (the one with the larger variance) to shrink its histogram and make it look like I_2 . Our aim is to maximize mutual information $H_M(\alpha I_1, I_2)$ with respect to α . Hence,

$$\frac{\partial}{\partial \alpha} H_M(\alpha I_1, I_2) = 0 \quad (3)$$

$$\frac{\partial}{\partial \alpha} \sum_{i_1} \sum_{i_2} p(\alpha I_1, I_2) \log_2 \frac{p(\alpha I_1, I_2)}{p(\alpha I_1) p(I_2)} = 0 \quad (4)$$

where $p(\alpha I_1, I_2)$ is the joint histogram of two images. If we go through the math from here, we would see that finding a 2D distribution as a function of I_1 and I_2 without loss of generality is quite difficult. Moreover, as Eqn. (5) shows, multivariate mutual information calculation is complicated.

$$H_M(I_1, \dots, I_N) = H_M(I_1, \dots, I_{N-1}) - H_M(I_1, \dots, I_{N-1} | I_N) \quad (5)$$

To mitigate the underlined complications, we utilize the mean square error (MSE) between feature maps. In [40], it is stated that there is a differential relation between mutual information and minimum MSE. The same reference [40] investigates input and output of transmission channels with Gaussian noise. Although the analysis is pertinent to a subject area, there are considerable similarities and they basically claim mutual information and MSE become very relevant.

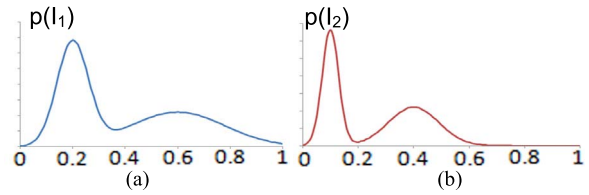


Fig. 6. Two sample histograms.

They derive the equations below which define the relation between MSE and mutual information.

$$\frac{d}{dSNR} H_M(X, Y) = \frac{1}{2} MSE(X|Y) \quad (6)$$

$$H_M(X, Y) = \frac{1}{2} \int_0^{snr} MSE(\gamma) d\gamma \quad (7)$$

What is claimed in this research is, for a specific scaling factor, the difference between two images to be fused needs to be minimal. Consequently, the redundancy between images should be maximized.

$$I_2 = \alpha^* I_1 + Error \quad (8)$$

In accordance with the approach in [40], given an input X a communication channel and an output Y , output will be some noise plus X scaled by signal to noise ratio (SNR).

$$Y = \sqrt{SNR} X + N \quad (9)$$

The two approaches are parallel in this aspect. Therefore, minimizing MSE between feature map histograms can generate similar results to mutual information approach.

$$MSE = \frac{1}{T} \int_{k=0}^1 [p_{\alpha I_1}(k) - p_{I_2}(k)]^2 dk \quad (10)$$

We can model probability distributions as Gaussian mixtures and set $T = 1$ since histograms are normalized:

$$MSE = \int_{k=0}^1 \left[\sum_i c_i \mathcal{N}(k | \alpha \mu_i, \alpha \sigma_i) - \sum_j d_j \mathcal{N}(k | \mu_j, \sigma_j) \right]^2 dk \quad (11)$$

Now, we minimize MSE w.r.t. α , subject to $\sum_i c_i = 1$ and $\sum_j d_j = 1$.

$$\frac{\partial}{\partial \alpha} \int_{k=0}^1 \left[\sum_i c_i \mathcal{N}(k | \alpha \mu_i, \alpha \sigma_i) - \sum_j d_j \mathcal{N}(k | \mu_j, \sigma_j) \right]^2 dk - \lambda_1 \frac{\partial}{\partial \alpha} (\sum_i c_i - 1) - \lambda_2 \frac{\partial}{\partial \alpha} (\sum_j d_j - 1) = 0 \quad (12)$$

Notice that the constraints are not dependent on α so they drop out. Also, note that the second Gaussian mixture is not dependent on α as well.

$$\int_{k=0}^1 \frac{\partial}{\partial \alpha} \left[\sum_i c_i \mathcal{N}(k | \alpha \mu_i, \alpha \sigma_i) - \sum_j d_j \mathcal{N}(k | \mu_j, \sigma_j) \right]^2 dk = 0 \quad (13)$$

Replacing Gaussian mixtures with the symbol G yields:

$$\int_{k=0}^1 \frac{\partial}{\partial \alpha} [G_1(k, \alpha) - G_2(k)]^2 dk = 0 \quad (14)$$

Eqn. (14) becomes (15) when chain rule is applied.

$$\int_{k=0}^1 2[G_1(k, \alpha) - G_2(k)] \frac{\partial}{\partial \alpha} G_1(k, \alpha) dk = 0 \quad (15)$$

This suggests two solutions:

$$[G_1(k, \alpha) - G_2(k)] = 0 \quad (16)$$

or

$$\int_{k=0}^1 \frac{\partial}{\partial \alpha} G_1(k, \alpha) dk = 0 \quad (17)$$

Eqn. (16) dictates that G_1 and G_2 are identical, which is highly unlikely. Solving Eqn. (17) is more likely to produce results. Eqn. (17) is sufficient for computational evaluation. However, it can be expanded to (18):

$$\int_{k=0}^1 \frac{\partial}{\partial \alpha} \sum_i c_i \frac{1}{\alpha \sigma_i \sqrt{2\pi}} \exp \left[-\frac{1}{2} \left(\frac{k - \alpha \mu_i}{\sigma_i} \right)^2 \right] dk = 0 \quad (18)$$

Eqn. (19) is obtained when (18) is solved for the partial derivative of α .

$$\int_{k=0}^1 \sum_i c_i \frac{-1}{\alpha^4 \sigma_i^3 \sqrt{2\pi}} \exp \left[-\frac{1}{2} \left(\frac{k - \alpha \mu_i}{\sigma_i} \right)^2 \right] \cdot \left(\alpha^2 \sigma_i^2 + \alpha k \mu_i - k^2 \right) dk = 0 \quad (19)$$

The first part of Eqn. (19) is a set of scaled Gaussians. Due to the negative sign it is negative for all values of α .

$$\frac{-c_i}{\alpha^4 \sigma_i^3 \sqrt{2\pi}} \exp \left[-0.5 \left(\frac{k - \alpha \mu_i}{\sigma_i} \right)^2 \right] < 0 \quad (20)$$

which leads to:

$$\int_{k=0}^1 \sum_i \left(\alpha^2 \sigma_i^2 + \alpha k \mu_i - k^2 \right) dk = 0 \quad (21)$$

This is an alternative to Eqn. (17). That is, Gaussians can be fitted to histogram data and α can be directly calculated. For a single Gaussian α evaluates to:

$$\alpha = \frac{\mu k \pm \sqrt{\mu^2 + 4\sigma^2}}{2\sigma} \quad (22)$$

The histograms of the features we extracted are indeed uni-modal almost all the experiments undertaken. By interpreting the mean and variance, α can be computed. Setting partial derivative w.r.t α to zero will give us the critical points; i.e., α^* . Local minima are negative to positive zero crossings of the first derivative at α^* location.

To demonstrate the feature fusion method, it is applied to two feature maps extracted from the sample frames given in Fig. 1. Figs. 7 and 8 depict the features used for fusion.

Since I_1 has greater variance, it is scaled with alpha coefficient to match I_2 histogram. The plot in Fig. 9 shows MSE of their histograms versus alpha:

Fig. 9 and Fig 10. show MSE and mutual information are closely related as it is claimed by the derivations. The peaks in HM vs. α plot correspond to the local minimums of the MSE vs. α plot. By selecting the minimum MSE generating

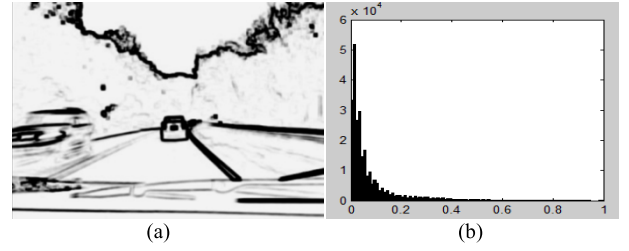


Fig. 7. Local variance of sample frame and its histogram taken as I_1 .

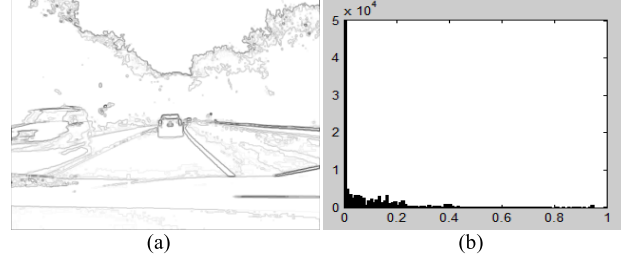


Fig. 8. Color cluster contours and its histogram taken as I_2 .

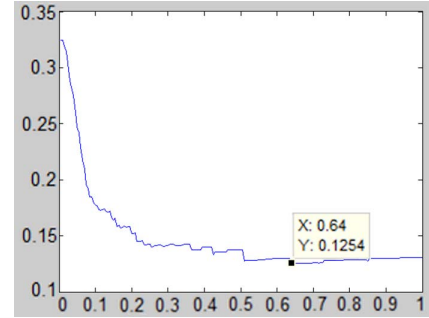


Fig. 9. $MSE\{p(\alpha I_1) | p(I_2)\}$ vs. α .

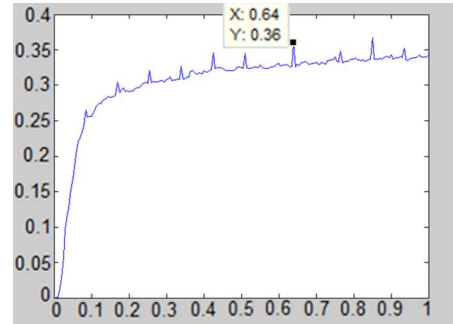


Fig. 10. $H_M(\alpha I_1, I_2)$ vs. α .

α we can find the optimal ratio that generates the highest information content.

The example demonstrates how the two images are combined by finding an optimal scaling coefficient. However, we combine more than two images. As mentioned earlier, this would complicate operational procedures if mutual information was used for fusion. With the minimum MSE approach, we keep the image with smaller variance constant and scale the intensity of the other image. By extending this approach, we can select the image with the minimum variance (among local variance, colors, optical flow, spectral energy, etc.) and find a scaling factor for the others.

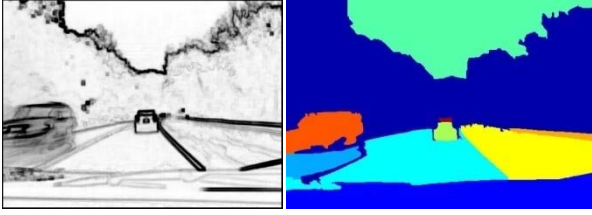


Fig. 11. Fused Saliency image and segmentation result after region separation and growing.

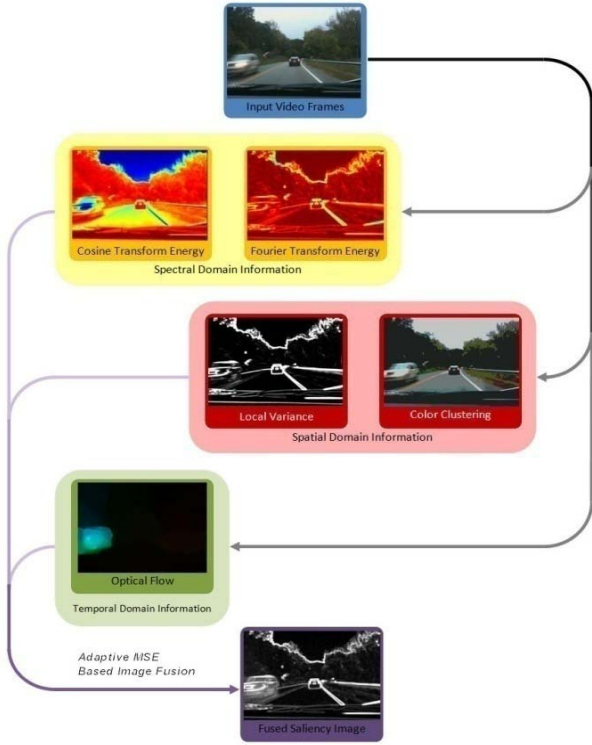


Fig. 12. Overall feature extraction and fusion.

This, in turn, gives a scaling rate (α^*) for each feature map and the scaling rate for the one with the minimum variance will be 1. The fused image can be given by

$$I_f = I_{\min_var} + \sum_i \alpha_i \cdot I_i \quad (23)$$

where $I_i \neq I_{\min_var}$.

Resulting fused image obtained from the combined saliency features is illustrated in Fig. 11. Fig.12 depicts overall feature extraction and fusion scheme. Notice that the colors of regions are assigned arbitrarily and do not signify any meaning.

III. RESULTS

A. Road Surface Detection

A set of experiments was carried out on a data set from [41]. Some video frames in this dataset are manually segmented to form a ground truth for validation. Although this sort of ground truth formation can be considered subjective, as long as subjects who do the segmentation abide certain set of rules based on their spontaneous vision, results will be consistent with small variations within statistical error margins. The ground

TABLE I
ROAD DETECTION ACCURACY RESULTS

Video Sequence	Road Detection Accuracy
005VD	0.9728
0006R0	0.9769
0016E5	0.9773
0001TP	0.9074

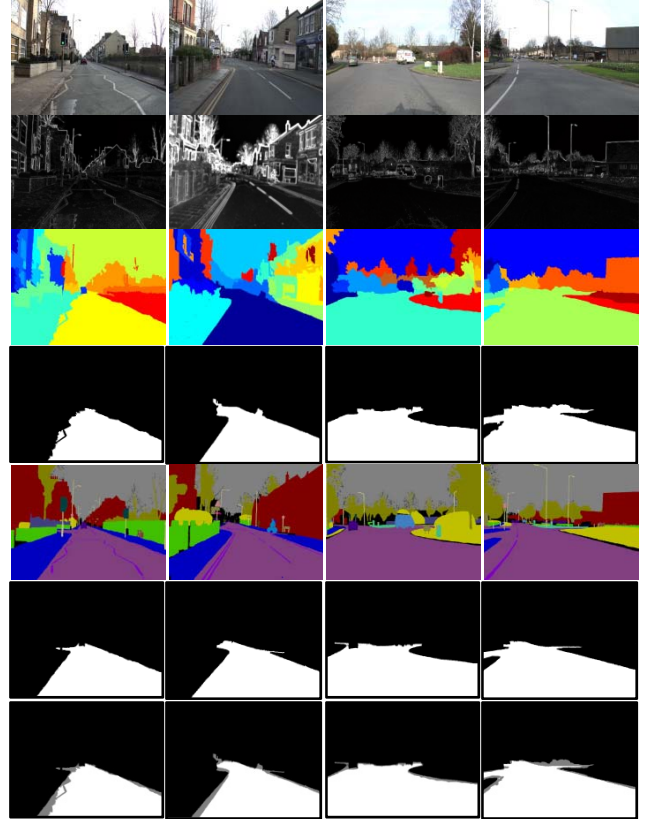


Fig. 13. Sample frames (row 1), their saliency (row 2), segmentation (row 3), road detection (row 4), ground truth (row 5), ground truth of road surface (row 6) and comparison of road detection to ground truth (row 7).

truth images are color coded. Purple tones indicate road and lane markings. Red regions are used for buildings, yellow is for vegetation, and gray is for the sky. By searching the purple colors on the ground truth images, road surfaces can be reached as the driving platform.

The images segmented with our method do not have such color coding. Colors only indicate label numbers which are arbitrary. The road surfaces are found by a maximum likelihood classifier using ontological features. These ontological features include properties such as shape, position and color. For instance, road is expected to be seen on the bottom half of the image. Therefore, position becomes an important indicator. Also, roads are in general straight or curved segments, warped by the visual perspective. Hence, we can use shapes as another important ontological feature for road surface detection.

Class models are constantly updated by new features. Emphasis of older features on class models is reduced over time in order to achieve adaptive learning.

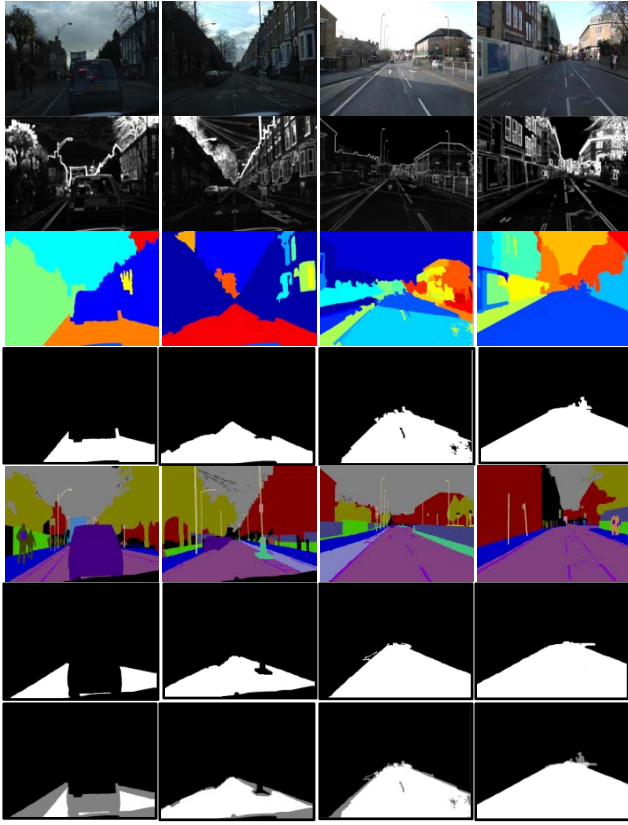


Fig. 14. Sample frames (row 1), their saliency (row 2), segmentation (row 3), road detection (row 4), ground truth (row 5), ground truth of road surface (row 6) and comparison of road detection to ground truth (row 7).

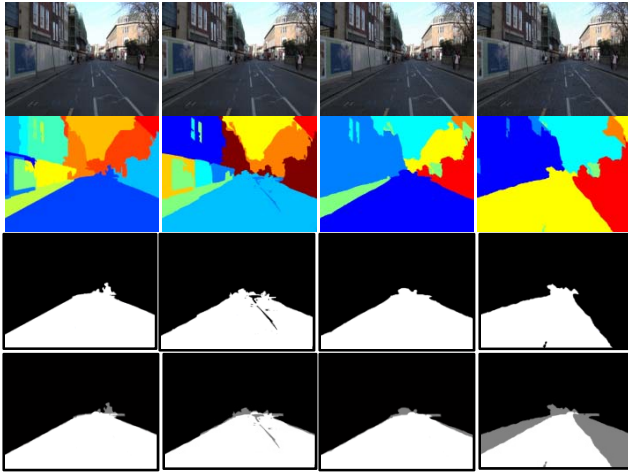


Fig. 15. Segmentation and road detection results of original image (1st column) and images degraded with Gaussian noise (images in columns 2, 3 and 4).

Figs. 13 and 14 demonstrate the results obtained from the aforementioned dataset, which contains four different video sequences. Two sample frames from each are shown here. The first row is the input scene. The second row is the saliency frame extracted with the new information fusion method. The 3rd row is the segmentation result. The 4th row contains the extracted road surface from segmented image. The next row shows the ground truth for the input frame. It is followed

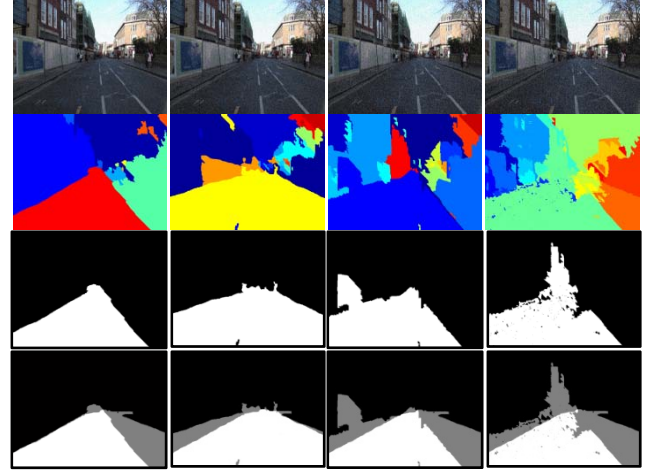


Fig. 16. Segmentation and road detection results of samples degraded with Gaussian noise (images in columns 2, 3, and 4).

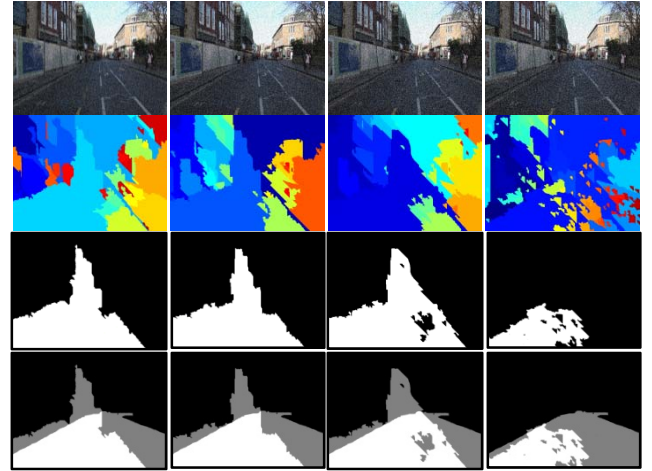


Fig. 17. Segmentation and road detection results of samples degraded with Gaussian noise (images in columns 2, 3, and 4).

by the road surface extracted from the ground truth. The last overlays road detection results on top of road surface ground truth images.

The accuracy of road detection is obtained by comparing the road elements in ground truth image (road and lane markers) to the extracted road surface from segmentation image. Measured accuracy is given by

$$DetectionAccuracy = \frac{1}{MN} \sum_{i=0}^{M-1} \sum_{j=0}^{N-1} f(i, j) \quad (24)$$

where M and N are image dimension and f is expressed as:

$$f(i, j) = \begin{cases} 1, & \text{if } I_R(i, j) = I_G(i, j) \\ 0, & \text{otherwise} \end{cases} \quad (25)$$

Here, I_R is the segmented road binary image and I_G is the ground truth for road surface. Accuracy measurements are taken for a series of frames in each video and averaged to give the overall accuracy. Table I summarizes the road surface extraction accuracy on four videos.

As seen in Figs. 13 and 14, the detected road surfaces (4th row) and ground truth (6th row) show great resemblance.

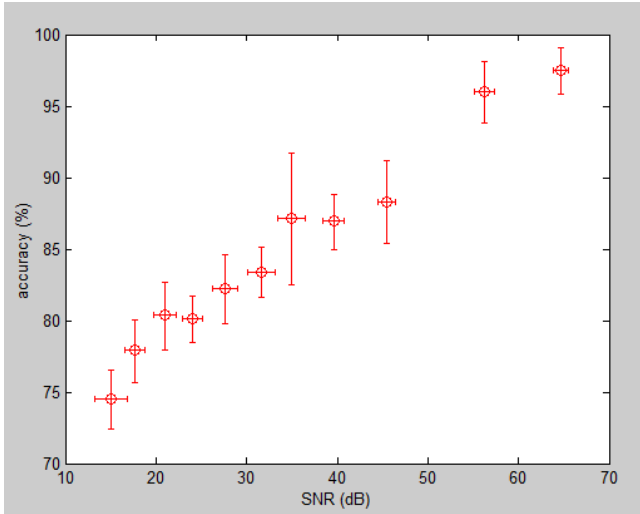


Fig. 18. Accuracy of road detections versus image SNR.

The last row demonstrates the segmentation results and ground truth overlaid. White regions are the parts marked as road on both segmented image and ground truth. Black regions are the non-road segments in both images. The gray parts show mismatches. The gray regions constitute a very small part of the entire overlaid image.

B. Noise Immunity Measurements

To measure how well the method performs under noisy conditions, we introduced pseudo Gaussian noise into a sample frame and measured segmentation accuracy with varying degrees of noise. Total of 11 copies of the original frame are created with progressively worsening noise. There is approximately 5 dB SNR drop on each example. Figs.15–17 display the frames in question and segmentation results along with the comparison to ground truth. The first image is the original frame, followed by noisy frames. The second row is the segmentation and the 3rd row is the road surface detection result. The last row compares road extraction to the ground truth.

Each test image except for the first one is degraded by a Gaussian white noise of varying strength. The signal to noise ratio (SNR) is calculated by

$$SNR = 10 \log_{10} \left(\frac{P_S}{P_N} \right) \quad (26)$$

where P_N and P_S represent noise and signal power, respectively. Signal power is given by the mean square of the image intensity.

$$P_S = \frac{1}{MN} \sum_{i=0}^{M-1} \sum_{j=0}^{N-1} I^2(i, j) \quad (27)$$

For a zero mean Gaussian noise, its power is given by its variance:

$$P_N = \text{var}(N) = \sigma_N^2 \quad (28)$$

Fig. 18 and Table II show the SNR of each noisy frame and the respective accuracy of road detection on these frames.

TABLE II
NOISE IMMUNITY RESULTS

Image sample 1		Image sample2		Image sample3		Image sample 4	
SNR	accuracy	SNR	accuracy	SNR	accuracy	SNR	accuracy
original	0.990	original	0.972	original	0.973	original	0.988
64.60	0.981	63.75	0.951	65.22	0.978	65.31	0.990
56.08	0.985	55.03	0.934	56.56	0.957	57.37	0.964
45.57	0.842	44.32	0.891	45.91	0.891	46.24	0.909
39.64	0.895	38.19	0.865	39.84	0.850	40.83	0.868
35.42	0.937	34.41	0.828	36.70	0.856	33.19	0.866
31.58	0.814	32.45	0.847	33.08	0.848	29.41	0.828
28.48	0.797	27.07	0.821	28.98	0.855	25.90	0.816
25.08	0.800	22.83	0.783	24.72	0.798	23.47	0.821
21.73	0.805	19.28	0.772	21.23	0.824	21.99	0.819
18.74	0.800	16.09	0.752	18.10	0.769	17.68	0.795
15.55	0.776	12.70	0.743	14.77	0.729	16.97	0.733

Image 1 and its reduced SNR versions listed in Table II are shown in Fig. 15 through Fig. 17. As the results indicate, even with a 50dB overall drop in SNR, accuracy decreases only by about 25%. There are some variations in accuracy due to random nature of noise. However, these fluctuations and overall drop in accuracy is small. Even when the segmentation suffers from extreme noise injected into the frames, the road surface extraction is still generating plausible perception results.

IV. CONCLUSION AND FUTURE WORK

The motivation behind this research is to cohesively integrate multiple approaches for an image context understanding problem effectively with robust performance. The proposed method stems from the human visual perception of a scene in its spatial, spectral and temporal attributes as a whole entity. To this end, instead of using one source, we can combine all of the available information yielding better results in mimicking human perception and cognition. In turn, this benefits computer vision and machine learning systems as demonstrated throughout the work presented herein. Forming an approach to a problem based on the conflicts and agreements of different methods can generate solutions that are much more general than solutions obtained with a single method. The research aims formation of a computer vision approach that produces higher level image descriptors by fusing low level image features together. These high-level descriptors can close the semantic gap in image understanding problems. Utilization of high level features lead to better segmentation, better tracking and eventually better scene understanding. Experimental results of this work indicate mimicking human vision to achieve a computer vision based driver assistance system produces very promising results. Presented results demonstrate the

effectiveness of combining multiple visual features together in order to acquire more informative features. As the ground truth comparisons indicate, the road surface detection accuracy is over 90%. Moreover, the proposed method performs well under noise, not losing much accuracy even after SNR drops of 50dB.

All experimentation in this research is carried out with daytime driving videos. As a future work, the algorithm presented here will be extended toward nighttime driving scenarios. Integration of infrared cameras can further improve the nighttime performance of the system. Another possible extension is the use of stereo cameras to obtain depth perception. Stereo vision can significantly increase the information content of the video frames.

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